MILITARY SPECIFICATION

PERISCOPE, TANK: M32CE1

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of combination passive-daylight tank gunner's periscope designated Periscope, Tank: M32CE1. The periscope is a part of the fire control system in the 165MM Gun, Combat Engineering Vehicle: M728 and is used for day and night. sighting and observing.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-E-48441	Elbow Assembly, Passive: 11733401
MIL-B-50645	Body (Daylight) Assembly:10516576
MIL-H-48620	Head Assembly: 11747439

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1240

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS (see 6.4)

US Army Armament Research and Development Command (ARRADCOM)

8247391	Test Fixture for Periscope
11732155	Gage, Interchangeability, Minimum
11732156	Gage, Interchangeability, Maximum
11738140	Final Acceptance Inspection Equipment
11747440	Periscope, Tank: M32CE1

PACKAGING DATA SHEET

11747440	Packaging of Periscope, Tank: M32CE1

(Copies of specifications, standards, drawings and packaging data sheet required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. -The periscope shall be manufactured in accordance with Drawing 11747440 and drawings pertaining thereto. The following shall have met their respective specifications (except that they need not be subjected to separate temperature vibration and shock tests) and be interehangeable within the requirements of this specification.

Elbow Assembly, Passive:	11733401	MIL-E-48441
Body (Daylight) Assembly:	10516576	MIL-B-50645
Head Assembly:	11747439	MIL-H-48620

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3.2 Interchangeability.

a. Modules. -The periscope shall meet the requirements of 3.8.2, 3.8.5, 3.8.6, 3.8.6.1 and 3.8.8 without need for realignment of components or selection of modules listed in 3.1. Adjustments, other than those specified herein, are prohibited.

b. Periscope with mount. -The periscope shall be capable of insertion into the Mount Periscope: M118E1 (gages are specified in lieu of mount in 4.6.7.2). After insertion, the periscope shall be capable of being firmly secured and subsequently removed.

3.3 General specification. -The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking

3.4 Environmental.

3.4.1 Storage temperature. -The periscope shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (730 ~ 180F) after having been exposed and thermally stabilized at temperatures of -80° and +160°F.

3.4.2 Operating temperatures. -The periscope shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at temperatures of -40° and +150°F.

3.5 Operability.

3.5.1 Elevation input coupling. -The running torque required to rotate the elevation input coupling through maximum elevation and depression, shall not exceed 6 pound-inches (lb-ins) at standard ambient temperature (73° ± 18°F) or 10 lb-ins at extreme temperatures of -40° and +150°F.

3.5.2 Diopter movement - Daylight and Passive. -The running torque required to rotate the daylight diopter movement through a minimum range of minus 3.0 to plus 3.0 diopters and the passive diopter movement through a minimum range of minus 4.0 to plus 4.0

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diopters shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.5.3 Focusing mechanism - Passive. -The running torque required to rotate Ring 11733436 through the range specified in 3.8.7 shall be within the-limits of 4.0 and 14.0 lb ins-when checked at standard ambient- temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 24.0 lb-ins.

3.5.4 Disengagement of boresight knob locks - Daylight and Passive. -The direct force required to disengage each boresight knob from lock position shall be within the limits of 2.0 and 5.0 pounds (lbs) when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.5.5 Rotation of boresight knobs - Daylight and Passive. -The running torque required to rotate each boresight knob through the travel range specified in 3.8.3.1.1 and 3.8.3.1.2 respectively, shall be within the limits of 0.5 and 5.0 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 8.0 lb-ins.

3.5.6 Shutter lever and on-off switch - Passive. -The torque required to rotate the shutter from the off to on position shall be within the limits of 4.5 to 6.5 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the torque shall not exceed 10 lb-ins.

3.5.7 Gain adjust and reticle brightness control - Passive. - This requirement shall be met at standard ambient temperature and the temperature conditions specified in 3.4.2. Knob MS91528-OEIB shall be manually operable through the full limits of the rheostat.

3.5.8 Focusing lock - Daylight. -This requirement shall be met with a 10 lb-ins torque applied to the locking screw on Housing 11727432. When checked at standard ambient temperature and the temperature conditions specified in 3.4.2, there shall be no rotation of Eccentric 8205635 when a 48 ounce-inches (oz-ins) torque is applied to the eccentric.

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3.5.9 Objective focusing mechanism - Daylight. -This requirement shall be met with the locking screw backed off a minimum of two full revolutions from the locked position established in 3.5.8. The running torque required to rotate Eccentric 8205635 through a minimum of one full revolution in each direction shall not exceed 30 oz-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 50 oz-ins.

3.6 Vibration/Shock.

3.6.1 Vibration "A". -This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing 11733401). The vibration input to the periscope shall be at the interface of the periscope and mount. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be- .5G ± 6%.

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .5G to 1.5G's ± 10%.

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleleration of 1.5G's ± 10%.

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5G's ± 10%.

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5G's to .5G ± 10%.

f. At 5 Hz, the acceleration shall be .5G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the periscope shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the periscope shall meet the requirements of 3.8.3.2, 3.8.4.1, 3.8.9, 3.8.2, 3.8.3.1.1, 3.8.3.1.2, 3.8.5, 3.8.6, 3.8.6.1, 3.8.7, 3.8.8, 3.2, 3.1, 3.7 and then each module (see 3.1) shall meet their own respective post-vibration test requirements.

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3.6.2 Vibration "B". -This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 + 6.0 VDC to Contact 11733497 (Reference Drawing 11733401.). The vibration input to the periscope shall be at the interface of the periscope and mount. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0G ± 10%.

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of LOG ± 10%.

c. During a frequency sweep of 500 to 5 Hz, matain constant acceleration of 1.0G ± 10%.

d. At 5 Hz, the acceleration shall be 1.0G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minutes ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes + 3 minutes). There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the periscope shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the periscope shall meet the requirements of 3.8.3.2, 3.8.4.1, 3.8.9, 3.8.2, 3.8.3.1.1, 3.8.3.1.2, 3.8.5, 3.8.6, 3.8.6.1, 3.8,7, 3.8.8, 3.2, 3.1, 3.7 and then each module (see 3.1) shall meet their own respective post-vibration test requirements.

3.6.3 Shock. -This requirement shall be met with the image intensifier- and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Ref. Dwg. 11733401). The shock input to the periscope shall be at the interface of the periscope and mount. The periscope shall be subjected to 24 cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in each direction along each of the Two mutually perpendicular horizontal axes (all four sides). The shock pulse shall have a half sine wave with a time duration of 3 .2 milli-seconds. The periscope shall also be subjected to six cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in one direction along the vertical axis (normal operating position of periscope only). The shock pulse shall have a

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half sine wave with a time duration of 18 + 3 mini-seconds. Subsequent to shock, there shall be no failure or flicker of light emission from the image tube nor the reticle projector lamp. The periscope shall also show no evidence of physical failure. Following the shock test, the periscope shall meet the requirements of 3.8.3.2, 3.8.4.1, 3.8.9, 3.8.2, 3.8.3.1.1. 3.8.3.1.2, 3.8.5, 3.8.6, 3.8.6.1, 3.8.7, 3.8.8, 3.2, 3.1, 3.7 and then each module (see 3.1) shall meet their own respective post-shock test requirements.

3.7 Cleanliness. -With the periscope disassembled, the three modular units shall meet the following requirements:

a. Defects. -There shall be no evidence of moisture, grease, fingerprints-conddensates, fractures and adhesive separations on any glass component when viewing through either end of each module (including the unity power system).

b. Dirt.

(1) Body assembly (Daylight). -When viewed through the eyepiece shall be no more than three dirt particles appearing within a 56 mil diameter central field of view. There shall be no more than three dirt particles appearing outside the specified field of view. The angular separation between dirt particles shall be at least 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution. The size of any particle shall not exceed the apparent width of the reticle line.

(2) Elbow assembly (Passive). -When viewing through the eyepiece with the image intensifier tube energized, there shall be no more than five particles visible on the screen surface. The size of any particle shall not exceed the apparent width of the reticle line.

(3) Head Assembly. -There shall be no more than three particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05 inch.

3.8 Performance.

3.8.1 Orientation.

a. Datum Plane “W” (Ref. Dwg. 11747441) defined by the three mounting pads in a vertical plane. Datum Plane "Y" (Ref. Dwg. 11747439) on a horizontal plane.

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b. Side "L" (Ref Dwg. 11747439) of the input coupling positioned 22 degrees 30 minutes with respect to Datum Plane “W” as measured in a vertical plane.

NOTE: At no time shall. the input coupling be allowed to strike hard against the travel stops.

c. The requirements of 3.8.3.1 to 3.8.3.2 inclusive, shall be met with:

(1) The eyepiece of the daylight system focused on the reticle.

(2) The eyepiece of the passive system focused on the screen of the image intensifier tube.

(3) A 1200 meter target provided for both the daylight and passive systems.

d. Focus passive elbow assembly by rotating objective focusing Ring 11733436 for best resolution of target (1200 meters) image.

3.8.2 Parallax. -Parallax between the reticle and the image of a target at 1200 meters shall not exceed 0.1 mil when adjusted by Eccentric 8205635 on the daylight body assembly.

3.8.3 Collimation. -The following collimation requirements shall be established prior to meeting the the vibration of 3.6:

a. An incident line of sight which is originally perpendicular within 0.2 mil to Datum Plane “W”.

b. The input coupling positioned in accordance with 3.8.lb.

3.8.3.1 Pre-Vibration. -Rotate daylight and passive boresight knobs to establish coincidence of each boresight cross with the incoming line of sight (see 3.8.3a) prior to meeting requirements of 3.6.

3.8.3.1.1 Daylight viewing area.

a. Elevation knob travel. -The boresight knob shall provide a minimum travel of 5.0 mil up and 8.0 mils down from the collimation setting of 3.8.3a.

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b. Deflection knob travel. -The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.8.3a.

c. Line of sight variation. -Variation in the line of sight in elevation, as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.8.3a, shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.8.3.1.2 Passive viewing area.

a. Elevation knob travel. -The boresight knob shall provide a minimum travel of 6.5 mils elevation and depression from the collimated setting of 3.8.3a.

b. Deflection knob travel. -The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.8.3a.

c. Line of sight variation. -Variation in the line of sight in elevation, as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.8.3a shall not exceed 0.3 mil (total variation not to exceed 0.6 mil).

3.8.3.2 Post-Vibration/Post-Shock Collimation. -The original collimation settings established in 3.8.3.1 shall not deviate more than 0.3 mil (including fixture error) as observed through the eyepiece subsequent to meeting requirements of 3.6.

3.8.4 Collimation repeatability.

3.8.4.1 Assembling daylight body and passive elbow to head assembly. -The daylight body and passive elbow assemblies shall be capable of repeating an original line of sight within 0.20 mil when removed and reinstalled (a minimum of 3 times) into the head assembly.

3.8.5 Image-tilt - daylight and passive. -The image of a target line which is originally parallel to Datum Plane “Y” shall emerge from the eyepiece to this same surface within 2.0 degrees of arc.

3.8.6 Parallelism of reticle and image, passive. -The image of a target line originally parallel to Datum Plane “Y” shall be parallel to the horizontal reticle line within 2.0 degrees of arc.

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3.8.6.1 Parallelism of reticle and image, daylight. -The image of a target line originally parallel to Datum Plane "Y" shall be parallel to the horizontal reticle line within 2.0 degrees of arc.

3.8.7 Objective focusing range.

3.8.7.1 Daylight. -Rotation of Eccentric 8205635 shall cause the objective assembly to travel a minimum of 0.010 inch in each direction from the position established in 3.8,2.

3.8.7.2 Passive. -Rotation of Ring 11733436 shall cause the objective-to bring targets from 50 meters to 3000 meters into focus.

3.8.8 Resolution.

3.8.8.1 Daylight. -The resolution shall be 10 seconds of arc or less in all four meridians. Eyepiece focus shift between any two between any two meridians shall not exceed 0.25 diopter.

3.8.8.2 Passive. -The resolution shall be one minute of arc or less in two meridians. Eyepiece focus shift between the meridians shall not exceed 0.25 diopter.

3.8.9 Electrical characteristics - passive.

3.8.9.1 Image intensifier tube. -The image intensifier tube shall be energized by the application of +24 ± 6.0 VDC to Contact 11733497 or application of 3 VDC ± 0.3 VDC to Connector 11733493 (Ref. Dwg. 11733401).

3,8.9.2 Image steadiness. -There shall be no visible flicker, no movement of the image nor autlible sparking when observed for one minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

3.8.9.3 Reticle brightness. -With a minimum of 18 and a maximum of 30 volts applied to the Contact 11733497, the reticle markings shall be visible when viewing through the eyepiece with the image intensifier tube energized (see 3.8.9.1). The reticle brightness shall be adjustable from off to full bright by adjusting the reticle brightness control knob MS91528-OE1B clockwise.

3.8.9.4 On-Off Switch. -The image intensifier tube shall shut down when +24 vDC + 6 vDC is removed from Contact 11733497 with the on-off switch in the onposition. The tube shall not be energized by reapplication of the 24 VDC unless the on-off switch is recycled from off to on.

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3.9 First Article Inspection. -This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. -The periscope shall meet the workmanship requirements of MIL-F-13926 and applicable specifications as cited on Drawing 11747440. The optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. -The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Classification of Inspections. -Unless otherwise specified herein or in the contract, the following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection.

4.3.1 Submission. -The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provision of 4.3.2. Unless otherwise specified, the sample for first article approval tests shall consist of three

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periscopes plus three each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The responsibility (Government or Contractor) for first article testing shall be as specified in the contract.

4.3.2 Inspections to be performed.

4.3.2.1 Periscopes. -First article periscopes shall be tested as specified herein for all the requirements of this specification, except 3,6.2.

4.3.2.2 SQAP items. -The items covered by SQAP shall be tested as specified thereon, for all characteristics delineated on the SOAP.

4.3.3 Rejection. -If any periscope or SQAP item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component or subassembly in the sample to comply with any of the stated requirements.

4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. -Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.4.2 Examination. -The examinations listed herein for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

4.4.2.1 Components and subassemblies. -All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the Supplementary Quality Assurance Provisions (SQAP) listed in the Technical Data Package {TDP). In the absence of SQAP’s the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4.2.2 Periscopes. -Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, Inspection Level II, and the sampling plans specified in Tables I, II, III, IV and V herein. Examinations and

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tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III, IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 100 Percent inspection. -The requirements and tests in Table I shall be inspected on a 100% basis.

TABLE I - CLASSIFICATION OF DEFECTS
(100 Percent Inspection)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL	NONE		
MAJOR:			
101.	Line of sight variation, daylight viewing area	3.8.3.1.1c	4.6.6.3.3
102.	Line of sight variation, passive viewing area	3.8.3.1.2c	4.6.6.4.3
103.	Resolution, daylight viewing area	3.8.8.1	4.6.6.8
104.	Resolution, passive viewing area	3.8.8.2	4.6.6.9
105.	Image intensifier tube	3.8.9.1	4.6.6.1.1
106.	Image steadiness	3.8.9.2	4.6.6.1.2
107.	Reticle brightness	3.8.9.3	4.6.6.1.3
108.	On-Off switch	3.8.9.4	4.6.6.1.4
109.	Interchangeability, periscope with mount	3.2b	4.6.7.2
MINOR:	None		

NOTE: The tests in Table I shall be conducted at a temperature of 73° + 18°F.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
110.	Operability, elevation input coupling	3.5.1	4.6.2.1
111.	Operability, diopter movement Daylight and Passive	3.5.2	4.6.2.2

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TABLE II - CLASSIFICATION OF DEFECTS - CONTINUED

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65% Defective			
112.	Operability, focusing mechanism - Passive	3.5.3	4.6.2.3
113.	Operability, disengagement of boresight knob locks - Daylight and Passive	3.5.4	4.6.2.4
114.	Operability, rotation of bore sight knob - Daylight and Passive	3.5.5	4.6.2.5
115.	Operability, shutter lever and on-off switch - Passive	3.5.6	4.6.2.6
116.	Operability, gain adjust and reticle brightness control - Passive	3.5.7	4.6.2.7
117.	Operability, focusing lock - Daylight	3.5.8	4.6.2.8
118.	Operability, objective focusing mechanism Daylight	3.5.9	4.6.2.9
119.	Collimation pre-vibration setting	3.8.3.1	4.6.3.1
120.	Vibration “B”	3.6.2	4.6.3.3
121.	Post-vibration examination	3.6.2	4.6.3.4
122.	Post-vibration collimation	3.8.3.2	4.6.3.5
123.	Cleanliness	3.7	4.6.5
124.	Parallax	3.8.2	4.6.6.2
125.	Elevation knob travel, daylight viewing area	3.8.3.1.1a	4.6.6.3.1
126.	Deflection knob travel, daylight viewing area	3.8.3.1.1b	4.6.6.3.2
127.	Elevation knob travel, passive viewing area	3.8.3.1.2a	4.6.6.4.1
128.	Deflection knob travel, passive viewing area	3.8.3.1.2b	4.6.6.4.2
129.	Collimation repeatability, assembling daylight body to head assembly	3.8.4.1	4.6.3.6.1
130.	Collimation repeatability, assembling passive elbow to head assembly	3.8.4.1	4.6.3.6.2
131.	Image tilt, daylight and passive	3.8.5	4.6.6.5
132.	Parallelism of reticle and image, passive	3.8.6	4.6.6.6

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TABLE II - CLASSIFICATION OF DEFECTS - CONTINUED

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
133.	Parallelism of reticle and image, daylight	3.8.6.1	4.6.6.6.1
134.	Objective focusing range, daylight	3.8.7.1	4.6.6.7..1
135.	Objective focusing range, passive	3.8.7.2	4.6.6.7.2
136.	Workmanship	3.10	Visual, Tactile
MINOR:	NONE		

NOTE: The tests in Table II shall be conducted at a temperature of 73° ± 18°F.

4.4.3 Acceptance and rejection. -Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Special Sampling.

4.5.1 General. -One periscope shall be randomly selected as a sample from each produced or from each month’s production, whichever occurs first. The sample shall meet the requirements and tests in Table III.

TABLE III - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings, Visual
302.	General Specification	3.2	MIL-F-13926

NOTE : The tests in Table III shall be conducted at a temperature of 73° ± 18°F.

4.5.2 Interchangeability. -Three periscopes that have met the requirements and tests in Tables I and II shall be selected at random as a special sample from each month’s production. Each sample shall then meet the requirement and test in Table IV.

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TABLE IV - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Interchangeability, Modules	3.2a	4.6.7.1

NOTE: The tests in Table IV shall be conducted at a temperature of 730 ± 18°F.

4.5.3 Environmental. -Three periscopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table V and shall then meet the requirements and tests in Tables I and II.

TABLE V - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (-80° and +160°F)	3.4.1	4.6.1.1
305.	Operating temperatures (-40° and +150°F)	3.4.2	4.6.1.2
306.	Operability, elevation input coupling	3.4.2, 3.5.1	4.6.1.2, 4.6.2.1
307.	operability, diopter movement, Daylight and Passive	3.4.2, 3.5.2	4.6.1.2, 4.6.2.2
308.	Operability, focusing mechanism Passive	3.4.2, 3.5.3	4.6.1.2, 4.6.2.3
309.	Operability, disengagement of boresight knob locks - Daylight and Passive	3.4.2, 3.5.4	4.6.1.2, 4.6.2.4
310.	Operability, rotation of boresight knob - Daylight and Passive	3.4.2, 3.5.5	4.6.1.2, 4.6.2.5
311.	Operability, shutter lever and on-off switch - Passive	3.4.2, 3.5.6	4.6.1.2, 4.6.2.6
312.	Operability, gain adjust and reticle brightness control - Passive	3.4.2, 3.5.7	4.6.1.2, 4.6.2.7
313.	Operability, focusing lock - Daylight	3.4.2, 3.5.8	4.6.1.2, 4.6.2.8
314.	Operability, objective focusing mechanism - Daylight	3.4.2, 3.5.9	4.6.1.2, 4.6.2.9

NOTE: The tests for characteristics 306 through 314 inclusive, shall be conducted at -40° and +150°F.

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4.5.4 Verification, follow-up. -Three periscopes shall be randomly selected as samples from production lots approximately at 25 percent, 50 percent and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table VI.

TABLE VI - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
315.	Collimation, Pre-Vibration setting	3.8.3.1	4.6.3.1
316.	Vibration "A"	3.6.1	4.6.3.2
317.	Post-Vibration examination	3.6.1	4.6.3.5
318.	Post-Vibration collimation	3.8.3.2	4.6.3.6
319.	Collimation, Pre-Shock Setting	3.8.3.1	4.6.3.1
320.	Shock	3.6.3	4.6.3.4
321.	Post-Shock examination	3.6.3	4.6.3.5
322.	Post-Shock, collimation	3.8.3.2	4.6.3.6

NOTE: The tests in Table VI shall be conducted at a temperature of 73° ± 18°F.

4.5.5 Failure of sample. -Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.4.3)

4.5.6 Inspection Equipment. -Requirements for inspection equipment to perform examinations and tests are specified herein and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection designs.

4.5.7 Government furnished inspection equipment. -Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.8 Contractor furnished inspection equipment.

4.5.8.1 Test equipment. -In conjunction with 4.5.6, the special inspection equipment in 4.5.8.1.1 shall be utilized in performing the applicable tests of 4.6.

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4.5.8.1.1 Performance. -Special testing equipment to test performance of the complete periscope shall conform to the requirements of Drawing 11738140 (Test Set-up, Periscope) and all drawings pertaining thereto.

4.6 Methods of Inspection.

NOTE: At no time during the following tests shall the input coupling be allowed to strike hard against the travel stops.

4.6.1 Environmental.

4.6.1.1 Storage temperature. -The test equipment utilized in this test shall be in accordance with the "Test Conditions" specified in the "General Requirements" of MIL-STD-810 and the conditions of 3.4.1. The periscope shall be placed in the test chamber and the temperature of the chamber reduced gradually (rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling test) to -80°F. The periscope shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see above rate of temperature change) to +160°F. The periscope shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (730 ±+ 18°F) and the periscope remove from the test chamber. The periscope shall then be subjected to the visual and audible examination of 4.6.3.5. Periscopes which have passed the above examination shall be subjected to the remaining tests specified in 4.6.1.2 through 4.6.7.2, inclusive. The periscope shall conform to the requirements of 3.4.1

NOTE: Temperature testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to temperature cycling (see 3.1). A record of all data necessary to determine compliance with required performance shall be available.

4.6.1.2 Operating temperatures. -The test equipment utilized in this test shall be in accordance with the "Test Conditions" specified in the "General Requirements" of MIL-STD-810 and the conditions of 3.4.2. The periscope shall be placed in the test chamber and the temperature of the chamber reduced gradually (rate of temperature

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change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling test) to -40°F. The periscope shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at -40°F, the periscope shall be subjected to the tests in 4.6.2 to determine compliance with the applicable requirements of 3.5, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see above rate of temperature change) to +150°F. The periscope shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the periscope shall again be subjected to the tests in 4.6.2 to determine compliance with the applicable requirements of 3.5, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see above rate of temperature change) to room ambient temperature (730 ± 18°F) and the periscope removed from the test chamber.

NOTE: Temperature testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to temperature cycling (see 3.1). A record of all data necessary to determine compliance with required performance shall be available.

4.6.2 Operability.

4.6.2.1 Elevation input coupling. -Standard torque measuring equipment capable of measuring the torque values specified in 3.5.1 shall be utilized for this test. An appropriate adapter to accommodate the coupling shall be used with the torque measuring equipment. The running torque required to rotate the elevation input coupling through its full range of travel is each direction shall not exceed the value specified in 3.5.1 for the temperatures indicated.

4.6.2.2 Diopter movement - Daylight and Passive. -Standard torque measuring equipment capable of measuring the torque values specified in 3.5.2 shall be utilized for this test. An appropriate adapter, to accommodate the diopter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter movement through the range specified in 3.5.2. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.5.2 for the temperatures indicated.

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4.6.2.3 Focusing mechanism - Passive. -Standard torque measuring equipment capable of measuring the torque values specified in 3.5.3 shall be utilized for this test. An appropriate adapter to accommodate Ring 11733436 shall be used with the torque measuring equipment. Torque shall be measured while rotating the ring through the range specified in 3.5.3. The running torque required to rotate Ring 11733436 shall conform to the applicable requirements of 3.5.3 for the temperatures indicated.

4.6.2.4 Disengagement of boresight knob locks - Daylight and Passive. -Standard measuring equipment capable of measuring the force values specified in 3.5.4 shall be utilized for this tests The force required to disengage each boresight knob from the lock position shall conform to the applicable requirements of 3.5.4 for the temperatures indicated.

4.6.2.5 Rotation of boresight knobs - Daylight and Passive. -Standard torque measuring equipment capable of measuring the torque values specified in 3.5.5 shall be utilized for this test. An appropriate adapter to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating each boresight knob through the range specified in 3.5.5. The running torque required to rotate each boresight knob shall conform to the applicable requirements of 3.5.5 for the temperature indicated.

4.6.2.6 Shutter lever and on-off switch - passive. -Standard torque measuring equipment capable of measuring the torque values specified in 3.5.6 shall be utilized for this test. The torque required to rotate the shutter lever from the off to on position and then back-to the off position shall conform to the applicable requirements of 3.5.6 for the temperatures indicated.

4.6.2.7 Gain adjust and reticle brightness control - Passive. -Manually operate the knob MS91528-OEIB. The knob shall meet the requirements specified in 3.5.7, at the temperatures indicated in 3.4.2 in addition to the standard ambient temperature (73° ± 18°F).

4.6.2.8 Focusing lock - Daylight. -A torque wrench capable of applying the specified torque values in 3.5.8, together with a standard type driver bit that properly fits the screw slot of the locking screw and eccentric, shall be utilized for performing this test. With the specified torque applied to the locking screw and while observing the eccentric, apply in a clockwise direction, the

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specified torque to the eccentric. Then, while still observing the eccentric, apply, in a counterclockwise direction, the specified torque to the eccentric. To conform to the requirements of 3.5.8, there shall be no rotation of the eccentric when observed during each torque application.

4.6.2.9 Objective focusing mechanism - Daylight. -Back off the locking screw of Housing 11727432 a minimum of two full revolutions from its locked position. Torque shall be measured while rotating the eccentric through the range specified in 3.5.9. The running torque required to rotate the eccentric shall conform to the applicable requirements of 3.5.9 for the temperatures indicated. Upon completion of this test, return the eccentric to the position in 3.8.2 and torque the locking screw as specified in 3.5.8. Periscopes which have passed the above tests of 4.6.2.1 through 4.6.2.9 inclusive, shall be subjected to the tests in 4.6.3.

4.6.3 Vibration/Shock (Periscope].

4.6.3.1 Collimation, Pre-Vibration/Pre-Shock setting. -This test and the following test of4.6.3.6 shall be performed with the periscope positioned and adjusted as in 4.6.6. Prior to performing these tests, disengage the index pin “D” from the index plate of the test fixture and place the 10 second level on Surface “C” of the leveling base. The bubble of the level shall be centered within its. graduation lines. If it is not, then further adjustment is required in accordance with the procedures prescribed in paragraphs 7 and 8 of the set-up instructions contained on Dwg. 8247391. With the index pin “D” disengaged from the index plate and surface “C” of the leveling base level, view through each periscope eyepiece (daylight and passive) and obtain coincidence of each periscope reticle boresight cross respectively, and the projected target image cross-lines by means of the boresight knobs (do not slip knob scales). When coincidence has been obtained, record the boresight knob scale settings for use in the test in 4.6.3.6.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original recorded settings throughout remainder of tests herein.

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4.6.3.2 Vibration "A". -Prior to performing this test, the periscope shall be subjected to the test procedure specified in 4.6.3.1 to determine compliance with 3.8.3. Subsequent to the test of 4.6.3.1, the periscope shall be vibrated in accordance with MIL-STD-810, Method 514.2, Procedure VIII, except the test level and frequency range shall be as specified in 3.6.1. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the periscope along three mutually perpendicular axes while the periscope is maintained in it's normal mounting or operating position (as depicted in the front view on Drawing 11747440). The periscope shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops during vibration. Prior to vibration, the image intensifier tube and the reticle projector lamp shall be energized in accordance with 3.6.1. After allowing 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece (passive). The periscope shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.6.1. During each vibration sweep cycle, the screen surface of the tube and the reticle pattern shall be monitored to assure that there is no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the vibration test, the periscope shall be subjected to the tests in 4.6.3.5 through 4.6.3.7.2 inclusive. Periscopes that have passed the above tests shall then be subjected to the tests specified in 4.6.6, 4.6.7, 4.6.4 and 4.6.5 respectively, to determine overall conformance with 3.6.1.

NOTE: Vibration testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to vibration (see 3.1). A record of all data (collimation, previbration settings, etc.) necessary to determine compliance with required performance shall be available.

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4.6.3.3 Vibration "B". -Prior to performing this test, the periscope shall be subjected to the test procedure specified in 4.6.3.1 to determine compliance with 3.8.3. Subsequent to the test of 4.6.3.1, the periscope shall be vibrated in accordance with MIL-STD-810, Method 514.2, Procedure VIII,except the test level and frequency range shall be as specified in 3.6.2. Vibration shall be applied along three mutually perpendicular axes of the periscope. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the periscope along three mutually perpendicular axes while the periscope is maintained in it's normal mounting or operating position (as depicted in the front view on Drawing 11747440). The periscope shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features of Mount Periscope: M118E1. The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops during vibration. Prior to vibration, the image intensifier tube and the reticle projector lamp shall be energized in accordance with 3.6.2. After allowing a 15 second tube stabil-" ization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece. The periscope shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.6.2. During each vibration sweep cycle, the screen surface of the tube and the reticle pattern shall be monitored to assure that there is no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the vibration test, the periscope shall be subjected to the tests in 4.6.3.5 through 4.6.3.7.2 inclusive. Periscopes that have passed the above tests shall then be subjected to the tests specified in 4.6.6, 4.6.7, 4.6.4 and 4.6.5 respectively, to determine overall conformance with 3.6.2.

NOTE: Vibration testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to vibration (see 3.1). A record of all data (collimation, pre-vibration settings, etc.) necessary to determine compliance with required performance shall be available.

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4.6.3.4 Shock. -Prior to performing this test, the periscope shall be subjected to the test procedure specified in 4.6.3.1 to determine compliance with 3.8.3. The shock equipment and test procedure shall be in accordance with MIL-STD-810, Method 516.2, Procedure I, except conditions such as "G" values, time duration and wave form shock pulse shall be as "specified in 3.6.3. The periscope shall be clamped on the shock equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. Prior to shock, the image intensifier tube and reticle projector lamp shall be energized in accordance with 3.6.3. The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops during the shock test. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece (passive). The periscope shall then be shocked in accordance with the test procedure specified above, except the application of shook along the vertical axis shall be in one direction only (normal operating position of the periscope). Subsequent to each shock, the screen surface of the tube and the reticle pattern shall be observed. During these observations, there shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the shock test, the periscope shall be subjected to the tests in 4.6.3.5 through 4.6.3.7.2 inclusive. Periscopes that have passed the above tests shall then be subjected to the tests specified in 4.6.6, 4.6.7, 4.6.4 and 4.6.5 respectively, to determine overall conformance with 3.6.3.

NOTE: Shock testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to shock (see 3.1). A record of all data (collimation, pre-shock settings, etc.) necessary to determine compliance with required performance shall be available.

4.6.3.5 Post-Vibration/Post-Shock examination. -Subsequent to vibration/shock test, the periscope shall be thoroughly examined for physical defects resulting from-poor workmanship and materials to determine compliance with 3.6, prior to being subjected to the remaining tests specified herein. A visual examination shall also be made for missing or damaged parts, glass chips, foreign matter

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and bonding (separations, chipping or rupture). An audible examination shall be made for loose internal optical and mechanical components. Any one of the possible defects above shall be cause to consider a periscope defective.

4.6.3.6 Post-Vibration/Post-Shock collimation. -Subsequent to the test in 4.6.3.5, the periscope shall again be subjected to the test procedure specified in 4.6.3.1. Any difference from the recorded boresight knob scale settings established in 4.6.3.1 shall not exceed the tolerance specified in 3.8.3.2, as measured on the boresight knob scales. Record the boresight knob scale settings for use in the tests in 4.6.3.7.

4.6.3.7 Collimation repeatability.

4.6.3.7.1 Assembling daylight body to head assemblY. -Perform this test subsequent to test in 4.6.3.6. Remove the daylight body from the head assembly. Repeated removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each installation for repeatability of the daylight collimation measurement recorded in 4.6.3.6. The periscope shall meet the requirements specified in 3.8.4.1, as measured on the bore-sight knob scales.

4.6.3.7.2 Assembling passive elbow to head assembly. -Perform this test subsequent to the test in 4.6.3.7.1. Remove the passive elbow from the head assembly. Repeated removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each installation for repeatability of the passive collimation setting recorded in 4.6.3.6. The periscope shall meet the requirements of 3.8.4.1, as measured on the boresight knob scales.

4.6.4 Modules. -Subsequent to the examination and tests on the periscope, the head, body and elbow assemblies shall be separated and separately subjected to the “post-vibration” requirements and tests specified in their respective module specifications identified in 3.1. Subsequent to meeting the respective module specification post-vibration requirements and tests, the head, body and elbow assemblies shall be examined and tested in accordance with 4.6.5 to determine conformance to cleanliness as specified in 3.7.

4.6.5 Cleanliness. -Each separate assembly (see 4.6.4) shall be subjected to the tests in 4.6.5.1, 4.6.5.2 and 4.6.5.3 respectively, to determine compliance with 3.7.

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4.6.5.1 Body assembly (daylight). -The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot condles. The body assembly (daylight) shall conform to the requirements of 3.7a and 3.7b(1).

4.6.5.2 Elbow assembly (passive). -The image intensifier tube shall be de-energized for this portion of the test. With a light source held close to the objective window and the eyelens, view through the glass components of the elbow assembly to determine compliance with the cleanliness requirements of 3.7a. Upon successful completion of this portion of the test, energize the image intensifier tube from a power source capable of supplying and maintaining 24 volts DC. The image tube shall be permitted a stabilization period of 15 seconds prior to performing the cleanliness test. Subsequent to the tube stabilization period, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the eyepiece diopter ring. The screen surface of the image tube shall then be visually inspected for cleanliness in accordance with 3.7b(2) and shall conform to the requirements specified therein. "Cold" and "Hot" spots On the screen surface of the image tube shall not be considered dirt.

4.6.5.3 Head assembly. -The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the exit window, inspection for dirt shall be made by viewing the illuminated field specified in 4.6.5.1. The head assembly shall conform to the requirements of 3.7a and 3.7b(3).

4.6.6 Performance. -To determine compliance with 3.8, the following tests shall be performed with the complete periscope positioned on the special test equipment specified in 4.5.2.3.1. Unless otherwise specified during these tests, the image intensifier tube and the reticle projector lamp shall be energized; the passive Focusing Ring 11733436 shall be adjusted for best focus of the target image after the diopter adjustment knob of the eyepiece is set for best focus of the phospher grain on the screen of the image tube; Eccentric (for adjusting parallax) shall be adjusted for minimum parallax between the target image and reticle and then

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locked after the diopter knob of the daylight eyepiece is set for best focus of the periscope reticle; the sector lever of test fixture shall be engaged in the zero degree elevation position on the index plate and the 10" collimator (80 field of view) shall be adjusted for 1200 meters. Light assembly "K" shall be removed when performing tests for daylight system. When performing tests for passive system, light assembly "K" shall be utilized. The image intensifier tube and reticle projector lamp shall be permitted a 15 second stabilization period, prior to conducting the performance tests.

4.6.6.1 Electrical characteristics - passive.

4.6.6.1.1 Image intensifier tube. -Apply +24 VDC ± 6 VDC to Contact 11733497 (Reference Drawing 11733401) and cycle the on-off switch from off to on, by means of-the shutter lever. After allowing a 15 second tube stabilization period, view through the passive eyepiece and observe the image tube screen. The screen surface of the image intensifier tube shall be illuminated. Upon satisfactory completion of the test, remove the 24 VDC ± 6 VDC from Connector 11733497 and apply 3 VDC ± 0.3 VDC to Connector 11733493 (Reference Drawing 11733401). Cycle the on-off switch from off to on. After allowing a 15 second tube stabilization period, view through the elbow assembly eyepiece. The screen surface of the tube shall be illuminated to determine overall compliance with 3.8.9.1.

CAUTION: When the on-off switch is in the "On" position and the input voltage to the regulator is interrupted, the tube cannot be energized by reapplication of the input voltage unless the on-off switch is cycled from-off to on.

4.6.6.1.2 Image steadiness. -Perform this test with the reticle projector lamp off. Position an observation telescope at the passive eyepiece of the periscope. The telescope shall contain a crossline reticle pattern and be capable of focusing on the screen of the image tube. With the aid of the telescope, view through the passive eyepiece and establish the telescope reticle crosslines coincident to the projected target image crosslines. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visual flicker, no movement of the image nor audible sparking as specified in 3.8.9.2. Image movement shall be determined by means of the target image with respect to the observation telescope reticle pattern. Sparking shall be determined by means of an audio examination. Upon completion of this test, energize the reticle projector lamp.

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4.6.6.1.3 Reticle brightness. -Perform this test concurrent with the test in 4.6.6.1.1. With each voltage specified in 3.8.9.3 applied to Contact 11733497 (Ref. Dwg. 11733401) and the image intensifier tube energized, rotate the reticle brightness control knob MS91528-OE1B to the off position. Then while viewing through the elbow assembly eyepiece, rotate the control knob clockwise throughout its range of movement. During this clockwise rotation of the control knob, the reticle brightness shall be adjustable from off to full bright and the reticle markings shall appear clearly defined at each voltage setting when observed through the elbow assembly eyepiece to determine compliance with 3.8.9.3.

4.6.6.1.4 On-off switch. -This test may be performed at the same time the test of 4.6.6.1.1 is performed. With the voltage specified in 4.6.6.1.1 applied to Contact 11733497, view through the elbow assembly eyepiece and observe the illuminated screen surface of the tube. Then while the on-off switch is in the on position, remove the voltage from the contact and reapply. During the above observation, the screen surface of the tube shall shut off when voltage is removed and shall not illuminate by reapplication of the voltage unless the on-off switch is recycled from off to on-to be in conformance with 3.8.9.4.

4.6;6.2 Parallax. -View through the daylight eyepiece and establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the-boresight knobs. While viewing through the eyepiece, parallax will be recognized by an apparent displacement of the intersecting point of the periscope reticle boresight cross with respect to the intersecting point of the projected target image crosslines when the observer's head is moved from side to side or up and down. Displacement between the periscope reticle and target image shall not exceed the tolerance specified in 3.8.2. Upon satisfactory completion of this test, perform the tests specified in 4.6.6.7.1 and 4.6.2.8.

4.6.6.2.1 Parallax adjustment. -If adjustment is necessary, a standard type screw driver that properly fits the screw slot of the locking screw and eccentric shall be utilized for performing this adjustment. The locking screw of Housing 11727432 shall be backed off a minimum of two revolutions from its locked position. Then while performing the test in 4.6.6.2, rotate the eccentric clockwise or counterclockwise until the parallax requirement specified therein has been met. When the adjustment is complete, apply the applicable torque value specified in 3.5.8 to the locking screw utilizing the torque wrench specified in 4.6.2.8.

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4.6.6.3 Daylight viewing area.

4.6.6.3.1 Elevation knob travel. -Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobss While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as specified in 3.8.3.1.1a as measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counter-clockwise direction. The boresight cross shall travel as specified in 3.8.3.1.1a as measured on the target.

4.6.6.3.2 Deflection knob travel. -Perform this test utilizing the procedure in 4.6.6.3.1, except the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.8.3.1.1b as measured on the target.

4.6.6.3.3 Line of sight variation. -Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.8.3.1.1c. Re-establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotate the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the values specified in 3.8.3.1.1c to determine overall conformance with the requirements specified therein.

4.6.6.4 Passive viewing area.

4.6.6.4.1 Elevation knob travel. -Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as

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specified in 3.8.3.1.2a as measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counter-clockwise direction. The boresight cross shall travel as specified in 3.8.3.1.2a, as measured on the target.

4.6.6.4.2 Deflection knob travel. -Perform this test utilizing the procedure in 4.6.4.4.1, except the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.8.3.1.2b as measured on the target.

4.6.6.4.3 Line of sight variation. -Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob, a variation in elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.8.3.1.2c. Re-establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotate the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the values specified in 3.8.3.1.2c to determine overall conformance with the requirements specified therein.

4.6.6.5 Image tilt - daylight and passive. -The daylight and passive optical systems of the periscope shall be inspected utilizing holder “T” containing telescope “J” with special reticle “L” (Ref. Dwg 11738140). The vertical reticle line of the telescope shall be aligned plumb when in the test position on a qualified level surface, With the aid of the auxiliary telescope, view through the passive eyepiece of the periscope and establish the telescopes reticle crosslines coincident to the target image cross-lines. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.8.5, as determined by the tolerance gates provided on the telescope reticle. Repeat the above procedure with the telescope positioned at the daylight eyepiece of the periscope. The images formed by the two optical systems (daylight and passive) of the periscope shall meet the requirements specified in 3.8.5.

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4.6.6.6 Parallelism of reticle and image - Passive. -Perform this test while viewing through the passive eyepiece of the periscope with the unaided eye. Position one extremity of the periscope horizontal reticle line coincident to the horizontal target image line. The periscope horizontal reticle line shall be parallel to the horizontal target line within the tolerance specified in 3.8.6, as determined by the tolerance gates provided on the tar get.

4.6.6.6.1 Parallelism of reticle and image - daylight. -Perform this test while viewing through the daylight eyepiece of the periscope with the unaided eye. Position one extremity of the 400 meter range line on the periscope reticle coincident to the horizontal target image line. The range line shall be parallel to the horizontal target line with the tolerance specified in 3.8.6.1, as determined by the tolerance gates provided on the target.

4.6.6.7 Objective focusing range.

4.6.6.7.1 Daylight. -This test shall be performed subsequent to the test in 4.6.6.2. A dial indicator, graduated in .001 of an inch, mounted on an appropriate stand, a supporting medium capable of receiving the dial indicator as well as the body assembly to be tested and the screw driver specified in 4.6.6.2.1 shall be utilized for this test. With the daylight body removed from the periscope and firmly secured on the supporting medium, position the dial indicator so that its contact point touches the face of the moveable objective tube of the daylight body and set the dial to read zero with the indicating hand. Back off the locking screw of Housing 11727432 a minimum of two full revolutions from its locked position. Starting from zero reading of the dial indicator, rotate the eccentric until the objective assembly has traveled the minimum distance specified in 3.8.7.1, as measured on the dial indicator. Return the indicating hand of the dial indicator to zero by means of the eccentric and repeat this travel in the opposite direction. There shall be sufficient travel in each direction to meet the minimum requirements specified in 3.8.7.1, as measured on the dial indicator. Upon completion of this test, reinstall the daylight body in periscope and repeat the parallax adjustment procedure specified in 4.6.6.2.1. The daylight body shall then be subjected to the test in 4.6.2.8.

4.6.6.7.2 Passive. -This portion of the test shall be performed with the 10” collimator adjusted for 50 meters. With the diopter ring adjusted for best focus of the phosphor grain on the screen of

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the image tube, rotate the focusing ring of the elbow assembly for best focus of target image to determine compliance with 3.8.7.2. Upon satisfactory completion of this portion of the test, the 10" collimator shall readjusted for infinity to determine compliance with the 3000 meters specified in 3.8.7.2. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, again rotate the focusing ring of the elbow assembly for best focus of the target image to determine overall compliance with 3.8.7.2.

4.6.6.8 Resolution - daylight. -The daylight optical system of the periscope shall be inspected for resolution utilizing a calibrated dioptometer with a magnification of at least three power. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter adjustment knob of the periscope daylight eyepiece to indicate zero diopter. Position the dioptometer at the periscope daylight eyepiece. While viewing through the dioptometer, adjust the objective scale of the dioptometer for best focus of the resolution pattern appearing in the fourth quadrant of the target reticle located in the 10″ Collimator (Ref. Dwg. 11738140 Test Set-up, Periscope). Any change in diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.8.8.1.

4.6.6.9 Resolution - passive. -This test shall be performed while viewing through the passive eyepiece with the unaided eye. Best focus of both meridians of resolution pattern appearing in the first quadrant of the target reticle located in the 10" Collimator (Ref. Dwg. 11738140 Test Set-Up Periscope), shall be obtained by adjusting the passive eyepiece diopter ring. Any change in the diopter readings by adjustment of the passive eyepiece diopter ring to obtain the detectable line structure and proper line count in each meridian shall be recorded, as measured on the ring scale. The difference between the readings for the meridians shall not exceed the value specified in 3.8.8.2.

4.6.7. Interchangeability.

4.6.7.1. Modules. -Each sample periscope (see 4.4.2) shall be disassembled (removal of passive elbow from head and daylight body from head) to permit interchange of the component modules. The modules shall be identified and interchanged as shown in Table VI.

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TABLE VI

Original Module Positions

	PERISCOPE 1	PERISCOPE 2	PERISCOPE 3
Head Assembly (HA)	HA-1	HA-2	HA-3
Body Assembly (DL) (Daylight	DL-1	DL-2	DL-3
Elbow Assembly (P) (Passive)	P-1	P-2	P-3

Interchanged Module Positions

	PERISCOPE 1	PERISCOPE 2	PERISCOPE 3
Head Assembly (HA)	HA- 1	HA-2	HA-3
Body Assembly (DL) (Daylight)	DL-2	DL-3	DL- 1
Elobw Assembly (P) (Passive)	P-3	P-1	P-2

Upon reassembly of the three interchanged periscopes, each periscope shall be subjected to the tests specified in 4.6.6.2, 4.6.6.5, 4.6.6.6, 4.6.6.6.1, 4.6.6.8 and 4.6.6.9 to determine compliance with 3.2a.

4.6.7.2 Periscope with mount. -This test shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings 11732156 and 11732155 respectively to determine compliance with 3.2b.

5. PACKAGING

5.1 Packaging Packing and Marking. -Packaging, packing and marking shall be in accordance with packaging Data Sheet 11747440. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. -This periscope is utilized as a day or night sighting and observation instrument. This capability is made possible by the use of a passive sighting system and a conventional

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periscope optical system encased in a single body assembly. The passive portion utilizes the 24 volt vehicle current source which is regulated to 2.7 volts, the required voltage to actuate the image intensifier tube.

6.2 Ordering data. -Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. Requirements for First Article Submission, Including data requirements for certificates of conformance for materials.

6.3 Inspection equipment design.

6.3.1 Government design. -Unless otherwise specified in the contract, all inspection equipment specified by drawing numbner in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

6.3.2 Contractor design. -The contractor shall design and supply inspection equipment compatible with the “Methods of Inspection” specified in 4.6 of this specification and with the “Test Conditions” specified in the “General requirements” of MIL-STD-810. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials and dimensions shall be so selected and controlled,as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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6.3.3 Submission of designs for approval. -Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-1, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of submittal

6.4 Drawings. -Drawings listed in Section 2 of this specification under the heading US Army Armament research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Input Coupling. -At no time shall the input coupling be allowed to strike hard against the travel stops.

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